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Implantation of hUCB-MSCs generates greater hyaline-type cartilage than microdrilling combined with high tibial osteotomy

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Abstract

Purpose: To compare the outcomes of treating large cartilage defects in knee osteoarthritis using human allogeneic umbilical cord blood-derived mesenchymal stem cell (hUCB-MSC) implantation or arthroscopic microdrilling as a supplementary cartilage regenerative procedure combined with high tibial osteotomy (HTO).

Methods: This 1-year prospective comparative study included 25 patients with large, near full-thickness cartilage defects (International Cartilage Repair Society grade ≥ IIIB) in the medial femoral condyles and varus malalignment. Defects were treated with hUCB-MSC implantation or arthroscopic microdrilling combined with HTO. The primary outcomes were pain visual analogue scale and International Knee Documentation Committee subjective scores at 12, 24 and 48 weeks. Secondary outcomes included arthroscopic, histological and magnetic resonance imaging assessments at 1 year.

Results: Fifteen and 10 patients were treated via hUCB-MSC implantation and microdrilling, respectively. Baseline demographics, limb alignment and clinical outcomes did not significantly differ between the groups. Cartilage defects and total restored areas were significantly larger in the hUCB-MSC group (7.2 ± 1.9 vs. $5.2 \pm 2.1\ cm^2$, $p = 0.023$; 4.5 ± 1.4 vs. $3.0 \pm 1.6\ cm^2$, $p = 0.035$). The proportion of moderate-to-strong positive type II collagen staining was significantly higher in the hUCB-MSC group compared to that in the microdrilled group (93.3% vs. 60%, respectively). Rigidity upon probing resembled that of normal cartilage tissue more in the hUCB-MSC group (86.7% vs. 50.0%, $p = 0.075$). Histological findings revealed a higher proportion of hyaline cartilage in the group with implanted hUCB-MSC ($p = 0.041$).

Conclusion: hUCB-MSC implantation showed comparable clinical outcomes to those of microdrilling as supplementary cartilage procedures combined with HTO in the short term, despite the significantly larger cartilage defect in the hUCB-MSC group. The repaired cartilage

Abbreviations: HTO, high tibial osteotomy; hUCB-MSC, human allogeneic umbilical cord blood-derived mesenchymal stem cell; ICRS, International Cartilage Repair Society; IHC, immunohistochemistry; IKDC, International Knee Documentation Committee; MFC, medial femoral condyle; MOCART, magnetic resonance imaging of cartilage repair tissue; MRI, magnetic resonance imaging; VAS, visual analogue scale; WBL ratio, weight-bearing line ratio.

after hUCB-MSC implantation showed greater hyaline-type cartilage with rigidity than that after microdrilling.

Level of Evidence: Level II, Prospective Comparative Cohort Study.

KEYWORDS

cartilage repair, high tibial osteotomy, knee osteoarthritis, marrow stimulation, mesenchymal stem cell

INTRODUCTION

Advanced osteoarthritis presents with diffuse, large cartilage defects, accompanied by various pathologic conditions such as meniscal degeneration, tears and peri-articular chronic inflammation [41]. Current cartilage regenerative procedures are better suited for treating localised, focal defects in relatively young patients than large, full-thickness cartilage defects in the osteoarthritic joints of older patients [7]. High tibial osteotomy (HTO) is currently applied to treat medial osteoarthritis in relatively young and active patients, with favourable clinical outcomes [5, 14, 15, 29, 30]. However, the potential for long-term deterioration and future conversion to total knee arthroplasty persists [10]. Hence, efforts are underway to improve long-term outcomes and prolong survival after HTO by incorporating supplementary cartilage procedures [18, 22, 23, 34, 45].

Human umbilical cord blood-derived mesenchymal stem cells (hUCB-MSCs) offer several advantages among the various sources of cartilage repair; these include noninvasive cell collection [32], high expansion capacity [49] and hypo-immunogenicity [21], rendering them suitable as allogeneic off-the-shelf products [33]. Furthermore, hUCB-MSCs theoretically overcome age-related concerns, such as the diminished reparative potential of autologous cells [46], which is particularly relevant in patients with osteoarthritis [37, 43]. Previous randomised clinical trials have demonstrated the effectiveness and safety of hUCB-MSC treatment for patients with conditions other than Kellgren–Lawrence (K–L) grade IV osteoarthritis and severe deformities [24, 32]. The indication for this treatment corresponds, to some extent, to that of HTO. Accordingly, hUCB-MSC implantation combined with HTO is being considered, demonstrating promising results [22, 34, 44, 45].

Marrow stimulation can effectively treat small and medium defects [20, 40] and supplement HTO procedures with improved cartilage regeneration outcomes [12, 16, 23]. Microdrilling, a recently introduced next-generation marrow stimulation, allows deeper and more subchondral perforations using a drill with a smaller diameter [40]. The clinical outcomes and the quality of repaired tissues are better for microdrilling than conventional microfracture using awls [3, 8, 35]. The application of microdrilling combined with HTO has not been reported, and whether the outcomes of hUCB-MSC implantation and the cost-effective yet advanced microdrilling differ have not been determined.

The present study aimed to compare the outcomes of treating large medial femoral condyle (MFC) lesions in knee osteoarthritis using hUCB-MSC implantation or arthroscopic microdrilling as a supplementary cartilage regenerative procedure combined with HTO. It was hypothesised that hUCB-MSC implantation would result in better clinical outcomes and cartilage tissue repair than that obtained with microdrilling.

METHODS

Study design and participants

This prospective comparative cohort study was conducted at a single hospital between 1 April 2020 and 31 July 2023. The inclusion criteria comprised age >50 years with persistent knee pain for >3 months despite undergoing conservative treatment with medication and physical therapy; moderate-to-severe degenerative knee osteoarthritis of the medial compartment and varus malalignment (K–L grade ≥ III on standing anteroposterior knee radiographs; mechanical tibiofemoral angle more than varus 5° on standing full-length lower extremity radiograph). The exclusion criteria comprised MFC cartilage defects that were < International Cartilage Repair Society (ICRS) grade III or defect size < 2 cm^2; ligament laxity grade ≥ II (Lachman, posterior drawer, varus/valgus stress test) or a history of cartilage surgery on the target knee joint within a year.

After enrolment, all participants underwent HTO, and the specific cartilage procedures were determined based on the participants' decisions. The study group received hUCB-MSC implantation, while the control group underwent microdrilling. Four patients in the control group were lost to follow-up at 48 weeks. One patient was excluded because of a change in the intraoperative surgical plan. The demographic characteristics of the patients and radiographic findings were comparable between the study and control

TABLE 1 Patient baseline characteristics: Demographic, radiographic, cartilage and meniscus assessments.[a]

Characteristics	Study group ($n = 15$)	Control group ($n = 10$)	p Value
Age, years	59.2 ± 5.4	57.9 ± 3.7	0.316
Sex			1.000
Male	3 (20)	2 (20)	
Female	12 (80)	8 (80)	
Body mass index, kg/m^2	26.5 ± 2.8	26.9 ± 1.4	0.183
Affected side			0.349
Right	7 (46.7)	2 (20)	
Left	8 (53.3)	8 (80)	
Preoperative K–L grade			0.802
Grade III	13 (86.7)	9 (90)	
Grade IV	2 (13.3)	1 (10)	
Preoperative mTFA	−5.5 ± 2.2	−6.6 ± 3.4	0.471
Preoperative WBL ratio	24.6 ± 8.6	19.9 ± 13.7	0.360
MFC defect size, cm^2	7.2 ± 2.0	5.2 ± 2.1	0.023*
MFC ICRS grade			0.836
Grade IIIB	0 (0)	0 (0)	
Grade IIIC	3 (20)	3 (30)	
Grade IVA	9 (60)	5 (50)	
Grade IVB	3 (20)	2 (20)	
Kissing lesion (medial, >ICRS 3b)	12 (80)	7 (70)	0.566
Combined TCH defect treatment	4 (26.6)	2 (20)	0.702
Meniscal procedure			0.555
None	3 (20)	3 (30)	
Partial meniscectomy	5 (33.3)	1 (10)	
Subtotal meniscectomy	5 (33.3)	5 (50)	
Root repair	2 (13.3)	1 (10)	
Meniscal extrusion, mm	4.7 ± 1.9	4.5 ± 1.7	0.934
Meniscal extrusion (>3 mm)	10 (66.7)	8 (80)	0.467
Meniscal insufficiency (meniscal extrusion + subtotal meniscectomy)	11 (73.3)	9 (90)	0.307

Abbreviations: ICRS, International Cartilage Repair Society; K–L grade, Kellgren–Lawrence grade; MFC, medial femoral condyle; mTFA, mechanical tibiofemoral angle; TCH, trochlear; WBL ratio, weight-bearing line ratio.

[a]Values are presented as means ± SD or number of patients (%).

*Statistical significance ($p < 0.05$).

groups (Table 1). Most profiles of cartilage defects and meniscal status did not differ between the groups (Table 1); however, cartilage defects were significantly larger in the study group ($p = 0.023$).

Cell preparation

The Korea Food and Drug Administration approved hUCB-MSCs (CARTISTEM®; Medipost) for treating cartilage lesions in January 2012. It comprises vials, each containing 1.5 mL (7.5×10^6 hUCB-MSCs) and 4% lyophilised hyaluronic acid hydrogel. During the mixing, 1.5 mL of the cells were drawn into a disposable sterile syringe and dropped directly into the freeze-dried sodium hyaluronate hydrogel. The mixture remained at room temperature for 30 min before being transferred to a syringe for implantation. The therapeutic dose was determined based on the preoperative size of the cartilage defects determined via magnetic resonance imaging (MRI). One vial of cells was used for cartilages with <3 cm^2 defects, whereas two vials were used for those with >3 cm^2 defects.

Surgical procedures and rehabilitation protocol

A single senior surgeon (S-H. K.) conducted all surgical procedures. Arthroscopic microdrilling was performed prior to HTO, whereas hUCB-MSC implantation was performed after HTO. The surgical steps for arthroscopic microdrilling were as follows: Extra portals were established when necessary to address large cartilage defects spanning the anterior to posterior MFC aspects. The direct accessibility, that is, the ability to approach near perpendicular to the defect for drilling through the portals, was checked using spinal needles before their creation (referred to as 'direct' portals). The MFC cartilage defects were prepared interchangeably using the anteromedial, anterolateral and direct portals as the working and viewing portals. An arthroscopic gouge was used to create vertical walls at the edges of the defects. The calcified cartilage layer was removed using curettes and an arthroscopic shaver. Subsequently, arthroscopic microdrilling was performed. Perpendicular subchondral bone perforations were created at 1–2 mm intervals to ensure nonoverlapping drill holes and prevent wall collapse (Figure 1e,f). A 1.5 mm drill bit (ECT Internal Fracture Fixation Drill Bits; Zimmer Biomet) was used, and a guide (B-IP-1512; Bioretec Ltd.) was used to delicately control the microdrilling. The depth of subchondral penetration was 13–15 mm.

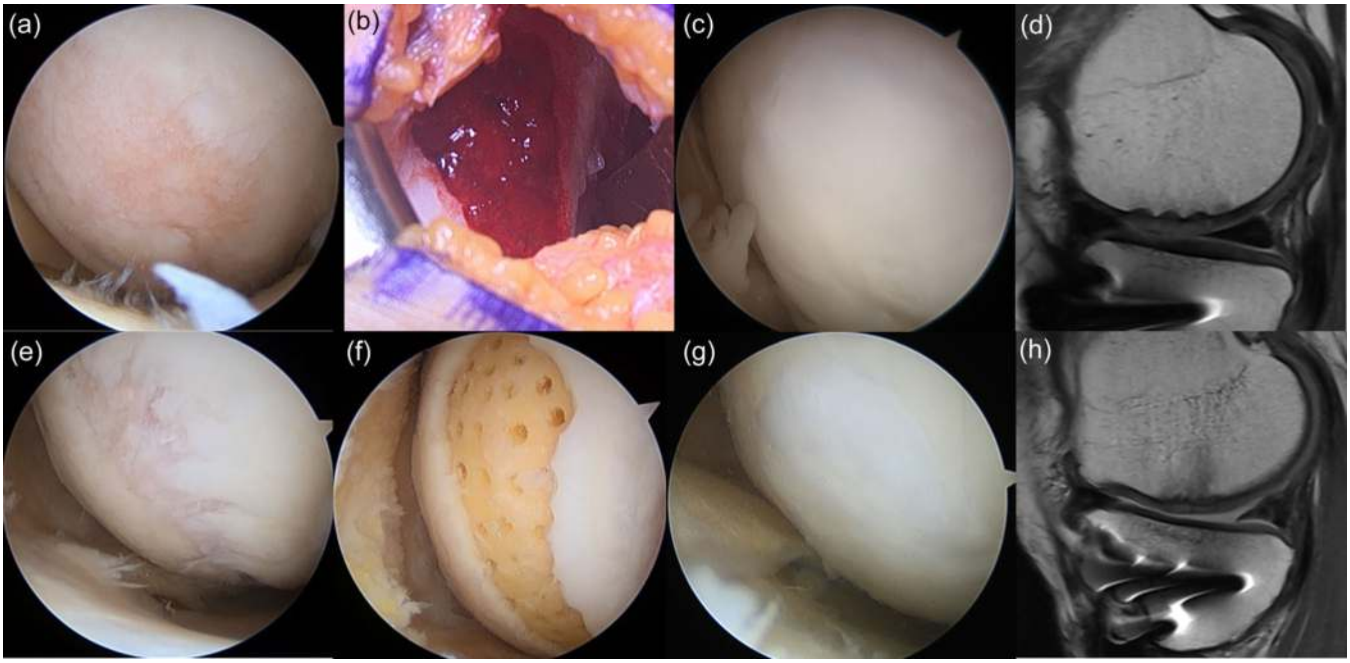

FIGURE 1 Implantation of human allogeneic umbilical cord blood-derived mesenchymal stem cells (hUCB-MSCs) and arthroscopic microdrilling. (a–d) Implantation of hUCB-MSCs. (e–h) Arthroscopic microdrilling. (a, e) Preoperative cartilage defect on medial femoral condyle viewed from anterolateral portal. (b) After hUCB-MSC implantation through a mini-arthrotomy. (c) Repaired cartilage after hUCB-MSC implantation on second-look arthroscopy. (d) Favourable cartilage regeneration on magnetic resonance imaging, except for subchondral bone oedema and minimal bony defect. (f) After arthroscopic microdrilling. (g) Repaired cartilage after microdrilling on second-look arthroscopy. (h) Favourable cartilage regeneration on magnetic resonance image, except for subchondral bone oedema.

The surgical procedure for the biplanar HTO was similar to that described previously [28]. Preoperative surgery was planned by realigning the mechanical axis to pass through the Fujisawa point, which represents 62.5% of the tibial plateau width from the medial edge. The osteotomy site was fixed using a TomoFix locking plate (Synthes).

hUCB-MSC implantations were performed after HTO. A mini-arthrotomy was performed in the parapatellar area through a longitudinal incision measuring 3–5 cm, which exposed lesions via further dissection (Figure 2a). Following cartilage lesion preparations by removing the damaged cartilage, multiple drill holes of two sizes ($4 \times 7\,mm^2$ and $2 \times 7\,mm^2$ [diameter × depth], spaced 1–2 mm apart) were created in the subchondral bone (Figure 2b and c). The mixture of hUCB-MSCs and hyaluronic acid was injected into the drill holes and then applied to cover the surface of the defect (Figure 2d).

Following the surgery, the knee motion was postoperatively restricted using a hinged knee brace for a total of 10 weeks. The patients were advised to start continuous passive range of motion exercises immediately after surgery, starting at 60° and increasing by 30° every 2 weeks. Weight-bearing was initially limited to toe-touch weight-bearing during the initial 4 weeks. The patients were permitted partial weight-bearing at approximately 50%, using crutches for support over the next 6 weeks. Following a comprehensive 10-week period, patients were encouraged to discontinue using crutches and resume normal walking.

Outcome assessments

Clinical outcomes were assessed using the pain visual analogue scale (VAS) and International Knee Documentation Committee (IKDC) subjective scores [1] at 12, 24 and 48 weeks, which is the primary outcome of this study. Secondary outcomes included arthroscopic, histological and MRI assessment, along with additional patient-reported outcome measures such as the Lysholm knee score [4] and Knee Injury and Osteoarthritis Outcome Score [36] at 1 year.

Arthroscopic assessments

The preoperative assessment involved a thorough arthroscopic evaluation of the ICRS grade and the size of MFC defects. A cartilage lesion rated ICRS grade ≥ IIIB on both sides of the tibiofemoral joint was defined as a kissing lesion. Plates were removed and cartilage regeneration was arthroscopically assessed approximately 1 year postoperatively. The repaired cartilage

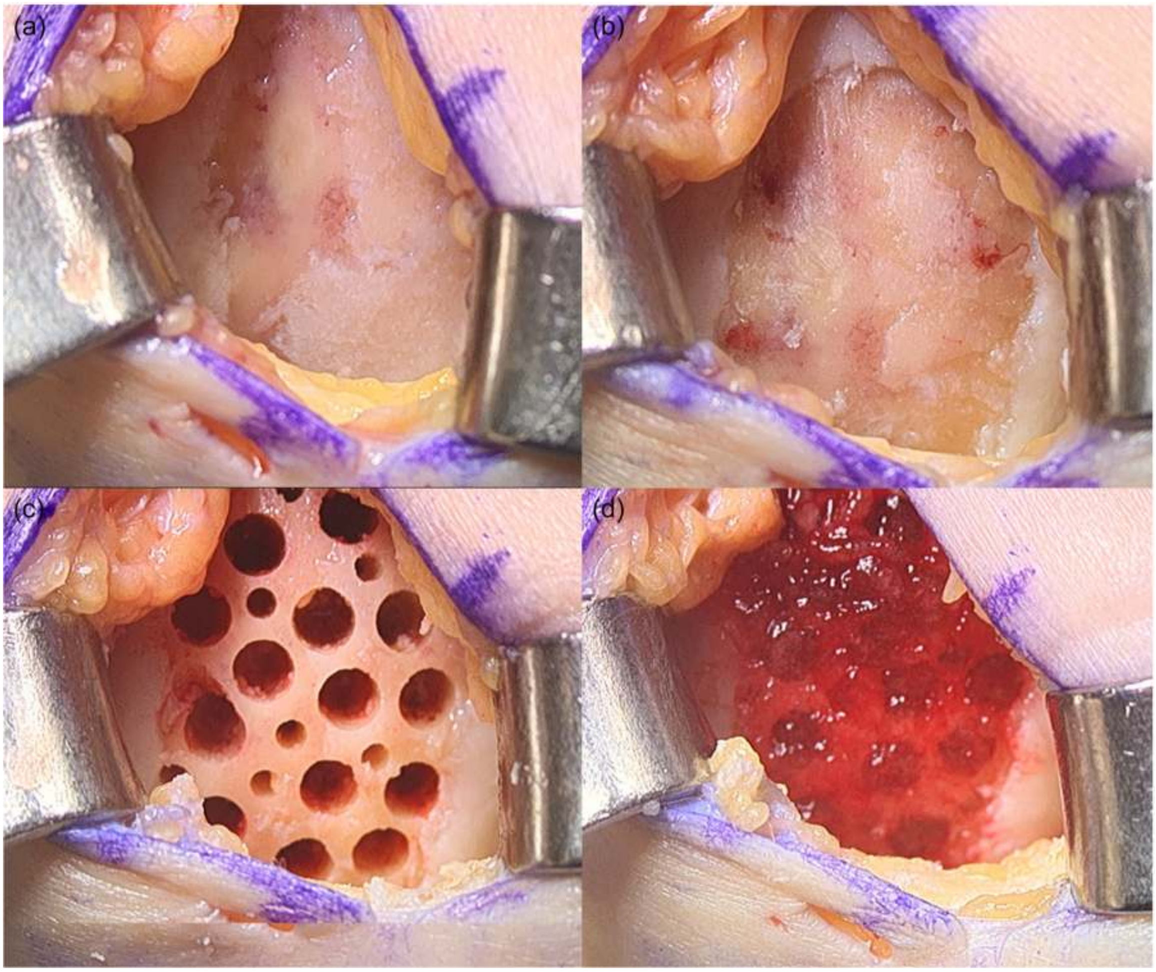

FIGURE 2 Process of the human allogeneic umbilical cord blood-derived mesenchymal stem cell (hUCB-MSC) implantation. (a) Exposure to the large cartilage defect on medial femoral condyle. (b) Defect preparation. (c) Subchondral drilling of the cartilage defect. (d) Implantation of the hUCB-MSCs.

was assessed using the ICRS cartilage repair assessment (CRA) score [31, 47], stratified as grades I (normal), II (nearly normal), III (abnormal) and IV (severely abnormal). Rigidity upon probing was categorised as rigid (normal compared with adjacent cartilage), softer and very soft compared with normal cartilage tissue, which is part of the Oswestry arthroscopy score system [47]. The proportions (%) of covered cartilage defects were calculated based on the initial sizes of cartilage defects and coverage. Outcomes were determined during arthroscopy via consensus among a senior orthopaedic surgeon and two orthopaedic fellows specialising in knee surgery and sports medicine. The final decision was reached through discussion among the investigators if there was a disagreement.

Radiographic and MRI assessments

Preoperative K–L grades were evaluated using standing knee anteroposterior radiographs. Pre- and postoperative alignment was evaluated as weight-bearing line (WBL) ratios (medial edge, 0%). Postoperative radiographic and MRI assessments were performed 1 day before the second-look arthroscopy and biopsies. The patients also provided informed consent to another postoperative MRI assessment using a 3.0-T Philips Ingenia (Philips Healthcare) approximately 1 year postoperatively. Two orthopaedic surgeons evaluated cartilage regeneration on MRI using the Magnetic Resonance Observation of Cartilage Repair Tissue (MOCART) 2.0 knee score [38]. Seven variables of the MOCART 2.0 knee score were also analysed to determine repaired tissue characteristics. Meniscal extrusion and the abundance of meniscal tissue were evaluated on MRI to determine meniscal function. Meniscal extrusion was measured using the previously described methods [27], and those >3 mm were considered pathological [6, 26]. Patients with pathological meniscal extrusion or a definite meniscal substance deficiency owing to abrasive destruction or subtotal meniscectomy during the procedure were defined as having 'meniscal insufficiency'.

Histological assessment

Cylindrical biopsies of repaired cartilage and subchondral bone were obtained during second-look arthroscopy and plate removal at approximately 48 weeks using a 13-gauge needle. Biopsies were fixed in 10% paraformaldehyde embedded in paraffin and cut into 3 μm-thick sections. General morphological features and glycosaminoglycan were visualised by staining the sections with haematoxylin and eosin (HE) and Safranin O, respectively. The sections were also immunohistochemically (IHC) stained using 1:200-diluted collagen types I (Ab34710; Abcam) and II (II-II6B3; Developmental Studies Hybridoma Bank) antibodies [13]. Two independent blinded pathologists assessed biopsy specimens using the ICRS II Histological Evaluation System [25]. Positive collagen type I/II IHC staining was visually categorised as strong, moderate and weak. According to overall histological assessments involving polarised light, HE and Safranin O staining, the repaired cartilage was categorised as predominantly hyaline (H), fibro-hyaline (F-H) or fibrocartilage (F). Hyaline cartilage is distinguished from fibrocartilage by a uniform matrix, particularly under polarised light. The cells within the hyaline cartilage are round or oval and are often surrounded by lacunae. In contrast, fibrocartilage consists of noticeable bundles of collagen fibres that are arranged randomly and irregularly under polarised light [19]. Samples in which >60% and 40%–60% of the matrix area comprised hyaline cartilage were categorised as H and F-H, respectively, whereas those in which >60% of the matrix area was fibrocartilage were categorised as F (Figure 3) [19].

Statistical analyses

The sample size was calculated using a two-sided superiority test with IKDC subjective scores comprising a significance level (α) of 5%, power ($1 - \beta$) of 90% and a possible dropout rate of 10%. The reference value (mean minimal clinically important difference) of 16.7 [11] and the standard deviation (SD) of 13 [39] for the IKDC subjective score were determined from the literature. Accordingly, the minimum sample size required for each group was 15. All data were statistically analysed using SPSS 26 (IBM Corp.), and values were considered statistically significant at $p < 0.05$. Values are presented as means ± SD unless otherwise indicated. For comparisons between the two groups, the Mann–Whitney U test was used for continuous variables, whereas Pearson's χ^2 or Fisher's exact test was used for categorical variables. Changes in primary outcomes within groups between before surgery and at every follow-up were assessed using repeated-measures analysis of variance. Post hoc analyses were conducted using Bonferroni correction. The intra- and inter-rater reliability of radiological and histological assessments were calculated as intraclass correlation coefficients (ICC). The assessments were conducted twice with a minimum interval of 3 weeks. The ICCs for intrarater reliability were 0.95 and 0.92 for MOCART and ICRS II scores, respectively, and those for inter-rater reliability were 0.89 and 0.91, respectively, indicating excellent reliability. Subgrouping into groups based on hyaline or fibrocartilage formation showed almost no variation in results between test–retest or between evaluators, demonstrating ICC value of 1.0 and 0.98 for intra- and inter-rater reliabilities.

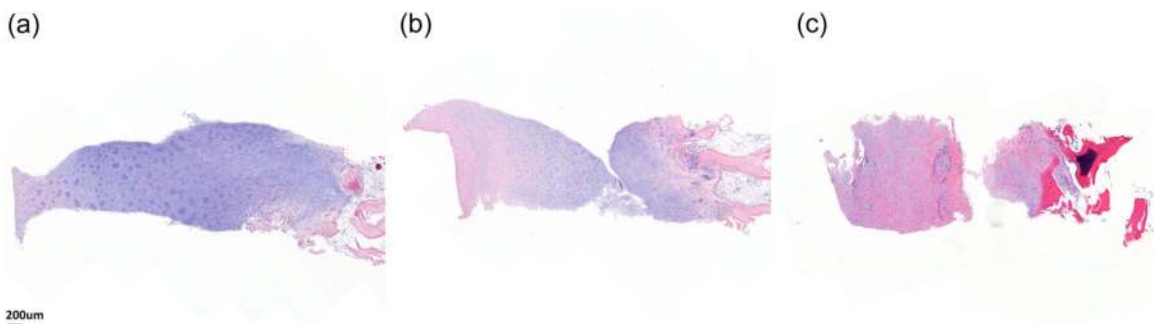

FIGURE 3 Subgrouping into three groups depending on hyaline or fibrocartilage formation. (a) Predominantly hyaline cartilage (H group) shows homogeneous basophilic material deposition in the mid/deep zone. (b) Fibrocartilage-hyaline mixture (F–H group) shows a heterogeneous mixture of basophilic and eosinophilic material deposition. (c) Fibrocartilage (F group) predominantly shows eosinophilic collagen deposition.

TABLE 2 Clinical outcomes and improved outcomes at 1-year postoperatively.

Outcomes	Study group	Control group	*p* Value
1-year clinical outcomes			
Visual analogue scale	17.9 ± 9.6	14.5 ± 14.6	0.292
IKDC subjective total	61.1 ± 11.7	58.7 ± 9.4	0.453
IKDC subjective symptom	22.6 ± 4.2	23.8 ± 3.3	0.737
IKDC subjective sports	24.0 ± 5.4	20.6 ± 5.1	0.164
IKDC subjective function	6.5 ± 2.5	6.7 ± 2.9	0.801
Lysholm knee score	73.1 ± 19.2	73.2 ± 12.8	0.781
KOOS total score	70.0 ± 10.8	71.1 ± 7.9	0.868
KOOS symptom	68.1 ± 13.7	69.6 ± 14.5	0.656
KOOS pain	77.2 ± 14.3	79.5 ± 10.6	0.739
KOOS ADL	79.3 ± 13.0	79.3 ± 8.1	0.956
KOOS sports	49.7 ± 16.8	42.5 ± 15.3	0.240
KOOS QOL	50.0 ± 20.9	55.6 ± 14.8	0.288
1-year improved outcomes			
Visual Analogue scale	35.9 ± 23.6	43.8 ± 30.5	0.328
IKDC subjective (total)	22.0 ± 12.2	23.4 ± 19.9	0.471
IKDC subjective (symptom)	7.3 ± 6.9	10.4 ± 9.3	0.220
IKDC subjective (sports)	8.5 ± 5.5	6.2 ± 6.7	0.676
IKDC subjective (function)	3.3 ± 2.4	3.2 ± 3.6	0.845
Lysholm score	19.1 ± 22.0	24.8 ± 23.5	0.420
KOOS total	16.1 ± 11.8	19.2 ± 20.6	0.698
KOOS symptom	8.1 ± 11.7	18.9 ± 14.5	0.359
KOOS pain	17.6 ± 15.4	25.6 ± 23.5	0.267
KOOS ADL	19.5 ± 14.4	17.4 ± 23.7	0.718
KOOS sports	20.0 ± 20.5	15.5 ± 21.4	0.636
KOOS QOL	14.6 ± 22.5	18.1 ± 16.5	0.503

Abbreviations: ADL, activities of daily living; IKDC, International Knee Documentation Committee; KOOS, Knee Injury and Osteoarthritis Outcome Score; QOL, quality of life.

TABLE 3 Cartilage regeneration assessment on second-look arthroscopy.[a]

Evaluation metrics	Study group ($n = 15$)	Control group ($n = 10$)	*p* Value
ICRS CRA overall points	8.5 ± 2.1	7.7 ± 3.4	0.800
Defect repair	3.7 ± 0.5	3.0 ± 1.4	0.263
Integration	2.1 ± 1.1	1.8 ± 1.0	0.578
Macroscopic appearance	2.8 ± 0.9	2.9 ± 1.3	0.707
ICRS CRA grade			0.327
Grade I (normal)	2 (13.3)	1 (10)	1.000
Grade II (near normal)	8 (53.3)	5 (50)	1.000
Grade III (abnormal)	5 (33.3)	2 (20)	0.659
Grade IV (severely abnormal)	0 (0)	2 (20)	0.150
Total coverage, %	68.0 ± 20.6	62.0 ± 27.8	0.445
Total covered area, cm^2	4.7 ± 1.5	3.0 ± 1.6	0.020*
Rigidity of repaired tissue			0.132
Rigid	13 (86.7)	5 (50)	0.075*
Softer	1 (6.7)	3 (30)	0.267
Very soft	1 (6.7)	2 (20)	0.543

Abbreviations: hUCB-MSCs, human umbilical cord blood-derived mesenchymal stem cells; ICRS CRA, International Cartilage Repair Society cartilage repair assessment.

[a]Values are presented as mean ± SD or number of patients (%).

*Statistically significant ($p < 0.05$) or borderline significance ($0.05 \leq p < 0.1$).

RESULTS

Clinical, arthroscopic and radiologic assessments

The VAS and IKDC subjective scores did not significantly differ between the groups at any time point (Table 2). Additionally, no significant differences in all patient-reported outcomes and improved outcomes at 1 year postoperatively also did not significantly differ between the groups (Table 2). On second-look arthroscopy, no significant differences were observed in all parameters assessed via arthroscopic evaluation using the ICRS CRA, including all subscores and grades (Table 3). Notably, no patients in the study group showed ICRS grade IV, whereas 20% of the control group did. The total coverage area was significantly

TABLE 4 Cartilage regeneration assessment on MRI.

Cartilage repair assessment	Study group	Control group	*p* Value
Total MOCART 2.0 score	50.3 ± 12.6	43.5 ± 16.2	0.261
Volume of cartilage defect filling	13.0 ± 5.3	11.5 ± 5.3	0.431
Integration into adjacent cartilage	6.0 ± 5.1	5.0 ± 5.3	0.601
Surface of the repair tissue	2.7 ± 2.6	3.5 ± 3.4	0.517
Structure of the repair tissue	0.7 ± 2.6	0.0 ± 0.0	0.414
Signal intensity of the repair tissue	9.3 ± 2.6	8.0 ± 4.2	0.325
Bony defect or bony overgrowth	7.0 ± 3.2	5.5 ± 3.7	0.299
Subchondral changes	11.7 ± 4.1	10.0 ± 5.8	0.542

Abbreviations: MOCART, magnetic resonance observation cartilage assessment of repaired tissue; MRI, magnetic resonance imaging.

larger in the study group (4.5 ± 1.4 vs. 3.0 ± 1.6, $p = 0.035$). The proportion of tissues evaluated as rigid was higher in the study group, with borderline significance (86.7% vs. 50%, $p = 0.075$). The study group had higher MOCART scores than did the control group, but the difference did not reach significance (Table 4). The WBL ratios were corrected from 24.6° ± 8.6° preoperatively to 66.4° ± 8.6° postoperatively in the study group and from 19.9° ± 13.7° to 68.3° ± 6.6° in the controls.

Histological assessment

Histological assessment using the ICRS II Histological Evaluation System did not find any significant differences between the groups (Table 5). Histological parameters for tissue morphology and matrix staining tended to be higher in the study group (tissue morphology, 82.0 ± 13.2 vs. 71.0 ± 22.8, $p = 0.189$; matrix staining, 85.3 ± 12.5 vs. 72.0 ± 18.7, $p = 0.061$). Mean overall ICRS II histological scores and positive IHC staining for collagen type II tended to be higher and more intense, respectively, in the study group, than in the microdrilling group, although the difference was not significant (Table 5). The proportion of patients in the H group was significantly higher in the study group than in the microdrilling group ($p = 0.041$) (Table 5 and Figure 4). Figure 5 shows the results of HE, Safranin O, and collagen type II IHC staining of regenerated tissue from implanted hUCB-MSCs in a patient.

TABLE 5 Histological assessment.[a]

ICRS II Histological Evaluation System score (0–100)	Study group	Control group	*p* Value
Overall	82.6 ± 12.8	76.0 ± 11.7	0.253
Tissue morphology	82.0 ± 13.2	71.0 ± 22.8	0.189
Matrix staining	85.3 ± 12.5	72.0 ± 18.7	0.061
Cell morphology	84.0 ± 19.2	83.0 ± 11.6	0.528
Chondrocyte clustering	92.7 ± 20.5	96.0 ± 5.2	0.586
Surface architecture	83.3 ± 15.0	90.0 ± 10.5	0.274
Basal integration	87.3 ± 22.5	83.0 ± 31.6	0.771
Tidemark formation	37.3 ± 42.8	41.0 ± 32.8	0.908
Subchondral bone abnormalities	86.7 ± 12.9	93.0 ± 10.6	0.187
Inflammation	100 ± 0.0	100 ± 0.0	1.000
Abnormal calcification	89.3 ± 19.8	92.0 ± 13.2	0.897
Vascularisation	94.0 ± 23.2	99.0 ± 3.2	0.814
Surface assessment	92.0 ± 11.5	94.0 ± 8.4	0.952
Mid/deep zone assessment	84.7 ± 13.6	71.0 ± 25.1	0.126
Immunohistochemistry staining positivity (collagen type II)			0.415
Strong	3 (20.0)	1 (10)	
Moderate	7 (46.7)	3 (30)	
Weak	5 (33.3)	6 (60)	
Overall tissue assessment			0.041*
Predominantly hyaline (H)	10 (66.7)	2 (20)	
Fibrocartilage-hyaline mixture (F-H)	4 (26.7)	4 (40)	
Fibrocartilage (F)	1 (6.7)	4 (40)	

Abbreviation: ICRS, International Cartilage Repair Society.
[a]Values are presented as mean ± SD or number of patients (%).
*Statistical significance ($p < 0.05$).

DISCUSSION

The primary findings of this prospective comparative study were that hUCB-MSC implantation as a supplementary cartilage procedure yielded comparable clinical outcomes despite significantly larger MFC cartilage defects. In addition, hUCB-MSC implantation yielded greater repaired hyaline-type cartilage than was obtained via microdrilling and a higher proportion of rigid cartilage compared with

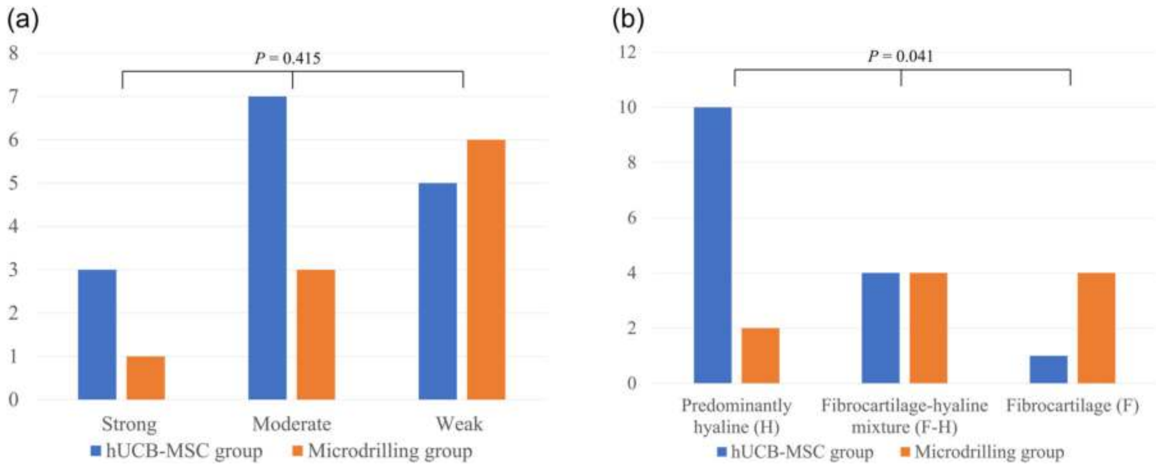

FIGURE 4 Proportions of (a) collagen type II positivity on immunohistochemistry stain and (b) overall tissue assessment categorised into H, F–H and F groups.

adjacent cartilage and more intense staining for type II collagen.

Treatment options are currently limited for restoring large cartilage defects, particularly among older patients with osteoarthritis. The commercial use of allogeneic hUCB-MSCs has emerged as a viable option with favourable outcomes [24, 42, 43]. A randomised controlled trial, extended to 5 years, found better clinical outcomes at mid-term follow-up after hUCB-MSC implantation compared with microfracture [24]. The proportion of patients with one or more improvements in ICRS grades at second-look arthroscopy (primary outcome measure) was significantly higher in the hUCB-MSC implanted group, providing further support for its efficacy. Several studies have also found that the outcomes of hUCB-MSC implantation were favourable when combined with HTO [18, 22, 34, 44, 45]. Despite the short follow-up in the present study, the mean IKDC score was comparable to those of previous studies (59.3–71.8). Arthroscopic outcomes have varied, with 64%–88% of patients being evaluated as normal or near normal according to the ICRS CRA. However, the arthroscopic outcomes were relatively poor in the present study, with a normal- to near-normal repaired cartilage proportion of 66.6%. This might be attributed to the significantly larger lesions in our study group, as the mean size of defects was notably larger than in those previously reported (7.2 ± 2.0 vs. 4.4 ± 2.1 [45] ~ 7.0 ± 1.9 cm^2 [22]). This could also be applied to the MRI outcomes. While MOCART scoring system is a valuable assessment tool, fully restored and well-integrated cartilage repair with excellent quality is given a perfect score of 100. Complete repair of large and diffuse lesions is inherently more challenging and inevitably leads to lower scores.

The hUCB-MSC implantation has been compared with several other procedures as a supplementary cartilage regenerative procedure combined with HTO [18, 22, 45]. Suh et al. [45] reported that the clinical outcomes and joint space widening after HTO were better after hUCB-MSC implantation than after microfracture. Lee et al. [22] reported a better ICRS grade with hUCB-MSC implantation on second-look arthroscopy than with bone marrow aspirate concentration implantation. Moreover, clinical outcomes and cartilage regeneration were found to be comparable to those after adipose-derived stromal vascular fraction implantations [18]. The present study could not find evidence to support the hypothesis that hUCB-MSC implantation would yield superior clinical outcomes compared with microdrilling. This might be attributed to the following aspects. The size of cartilage defects significantly differed between the groups. A large cartilage defect is considered a less favourable prognostic factor for cartilage restoration [17, 48]. Another reason was the short follow-up. A randomised control trial found no difference between the groups during the first post-operative year [24]. Another reason is that the clinical outcomes of cartilage procedures combined with HTO might be masked by the HTO procedure itself in the short term. The outcome of the regenerated cartilage is expected to be related to long-term outcomes or survival rather than short-term clinical outcomes [9, 39]. While our results did not clearly demonstrate superiority, hUCB-MSC implantation achieved results in all patient-reported outcomes comparable to those obtained via microdrilling, despite considerably larger initial cartilage defects. Additionally, even with a large and diffuse lesion, the group with the implanted hUCB-MSC had a similar coverage proportion relative to the total defect area. Therefore, our findings indicate that hUCB-MSC shows promise as an approach to cartilage procedures in patients with osteoarthritis with large and diffuse lesions.

Most secondary outcomes assessed by MRI and arthroscopy did not significantly differ between the

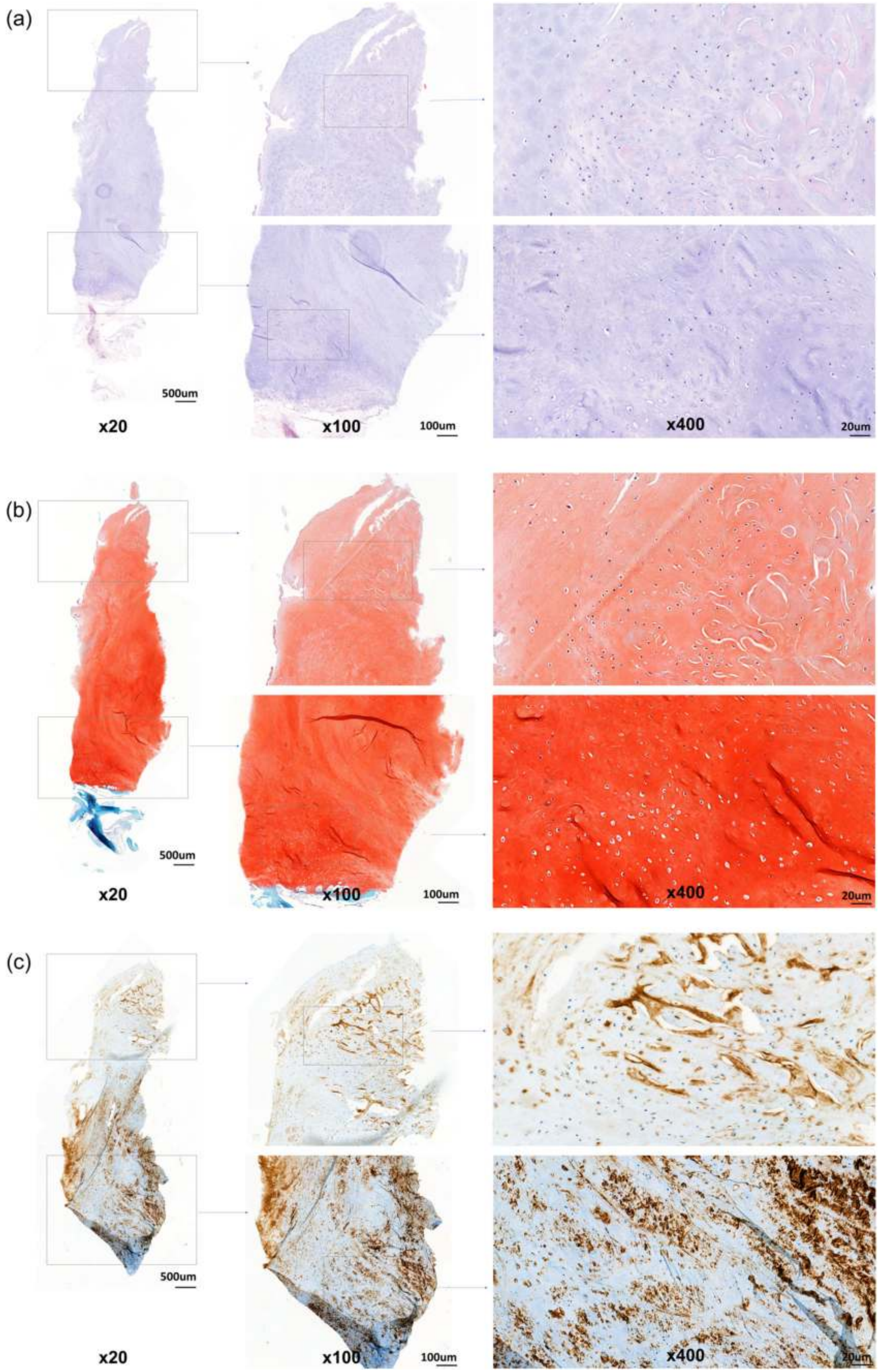

FIGURE 5 (See caption on next page).

groups. However, the rigidity of repaired tissue on probing during second-look arthroscopy was notably different between the groups. This assessment is part of the Oswestry arthroscopy score, a reliable system that is used to assess macroscopic cartilage repair [47]. Most of our patients in the hUCB-MSC group had adequate repair tissue rigidity, indicating good tissue quality. Additionally, 20% of patients in the microdrilling group had severely abnormal findings with no signs of cartilage regeneration, whereas the study group did not have such results.

One of the strengths of this study was the histological assessment, which is challenging without prospective planning. It considerably helped to discover some trends in the histological differences between the two groups. Although the ICRS II histological assessment did not reveal any statistically significant differences between the two groups, the overall assessment score was higher in the group implanted with hUCB-MSCs. Mean scores for tissue morphology and matrix staining parameters were also higher in the hUCB-MSC group, although a statistical difference was not achieved. Given the small sample, we cannot rule out the possibility of a type II statistical error. Tissue morphology parameters assess the distribution of collagen fibres, which is related to the tissue type (hyaline, fibrocartilaginous or fibrous) [25]. The matrix staining parameter indicates the proteoglycan content, which reflects the load-bearing qualities of the cartilage [2, 25]. Upon reviewing the type I and II collagen stains on the IHC, the results of the type I collagen stain were unsatisfactory. Type I collagen staining was weakly positive in all patients in both groups, which was related to the high specificity of the antibodies used in this study. However, satisfactory staining was observed for type II collagen staining. The hUCB-MSC group had a higher proportion of moderate-to-strong positivity for type II collagen on IHC staining than did the microdrilling group. In the overall tissue assessment, the proportion of patients whose repaired cartilage showed predominantly hyaline cartilage was significantly higher in the hUCB-MSC group. Overall, histological assessments in the present study demonstrated that hUCB-MSC implantation yielded greater hyaline-type cartilage than that with microdrilling.

This study had some limitations. First, the sample size was small, which was a major limitation. Obtaining patient consent for a biopsy in a clinical setting presented challenges. Concerns regarding the potential prolongation of the study period led to an initial decision to keep the sample size to a minimum. Unfortunately, five patients in the control group were lost to follow-up, which increased the risk of type II error in the statistical analyses and increased the possibility of selection bias. Second, the follow-up period was short. It was remarkably short to analyse differences in patient-reported outcomes after each cartilage procedure performed along with HTO. Further long-term follow-up studies are required to compare the efficacy of each procedure. Third, the second-look arthroscopic evaluation could not be blinded because of differences in postoperative scars (mini-arthrotomy vs. arthroscopy). Finally, cartilage defect size matching should have been considered before enrolment. Owing to significant differences in the mean defect size, this study did not detect differences in the primary outcome and most secondary outcomes between the two groups.

CONCLUSION

hUCB-MSC implantation showed comparable clinical outcomes to those of microdrilling as supplementary cartilage procedures combined with HTO in the short term, despite the significantly larger cartilage defect in the hUCB-MSC group. The repaired cartilage after hUCB-MSC implantation showed greater hyaline-type cartilage with rigidity than that after microdrilling.

AUTHOR CONTRIBUTIONS

The concept of the study was generated and the project was coordinated by Sung-Hwan Kim (corresponding author). Se-Han Jung (first author) drafted the manuscript together with Hyunjin Park, Chong-Hyuk Choi, Min Jung, Kwangho Chung and Hyun-Soo Moon. The histological assessment (blinded) and the description of the histological findings were performed by Hyunjin Park, who is a professional pathologist. The acquisition of data and analysis were done by Se-Han Jung, Jisoo Park and Ju-Hyung Lee. Sungjun Kim, who specialised in musculoskeletal radiology, participated in the acquisition and analysis of the radiologic data. Se-Han Jung and Sung-Hwan Kim revised the final draft critically for important intellectual content and approved the version to be submitted.

FIGURE 5 Histological findings from a patient that underwent human allogeneic umbilical cord blood-derived mesenchymal stem cell implantation show favourable cartilage repair. (a) Haematoxylin and eosin stain shows mixed eosinophilic and basophilic material deposition in the surface zone and homogenous basophilic material deposition in the mid and deep zones. All the layers show the chondrocytes with round to oval nuclei and surrounded lacunae. (b) Safranin O staining shows rich glycosaminoglycan, the main component of hyaline cartilage, in the mid and deep zones. (c) Collagen type II immunohistochemical staining shows a large amount of collagen type II deposition in the whole layer (×20, ×100 and ×400).

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflict of interest.

ETHICS STATEMENT

The institutional review board at Gangnam Severance Hospital approved the study protocol (Approval no: 2019-0502-011), and all participants provided written informed consent. The study complied with the approved protocol, the current version of Good Clinical Practices (IHC-E6), and the principles enshrined in the Declaration of Helsinki (2013 amendment).

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